INTRODUCTION TO DATA STRUCTURES

MIT SCHOOL OF COMPUTING

AS PART OF CURRICULAM

FIRST YEAR (SEM-2)

COMPUTER SCIENCE AND ENGINEERING

UNDER THE GUIDANCE OF

Prof. Dr. Sachin Tiwari

FOR THE ACADEMIC YEAR 2025-26

SUBMITTED BY

Aditya Satish Lavhande

Subject Teacher	**Course Coordinator**
Prof. Dr. Sachin Tiwari	Prof. Dr. Sachin Tiwari

EXPERIMENT DETAILS

Experiment Number	Experiment Name
1	Arithmetic Operations
2	Built-in Functions
3	Loops
4	Data types
5	String

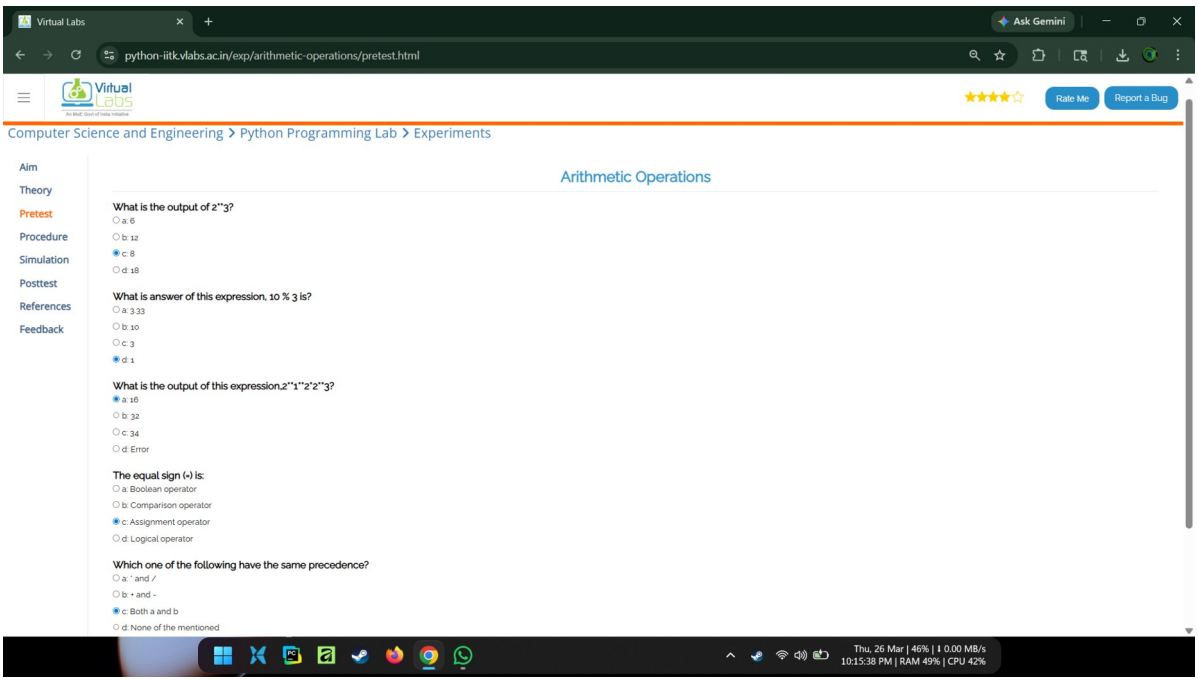
Virtual Lab Name: Virtual Labs

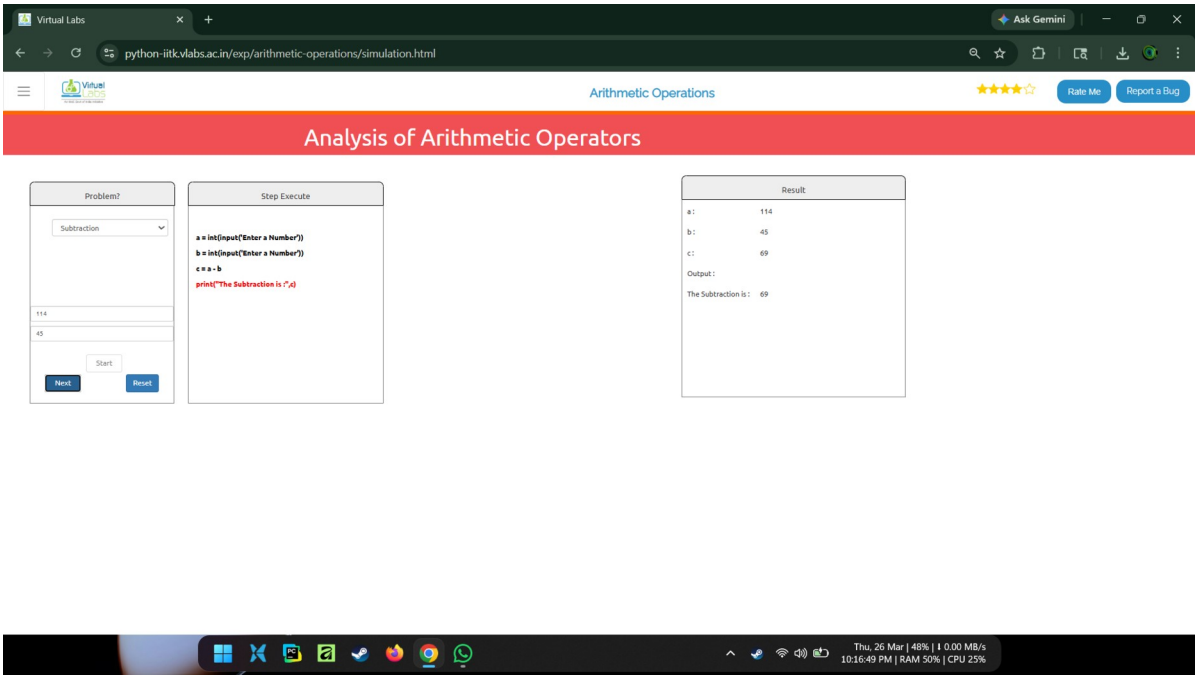
Institute: IIIT-H(Hyderabad)

Date of Performance: 26 March 2026

SCREENSHOTS

Arithmetic Operations





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- Aim
- Theory
- Pretest
- Procedure
- Simulation
- Posttest
- References
- Feedback

Arithmetic Operations

Is Python a case sensitive language?

- a. No
- b. Yes

What will be the output of the following code?

```
x= 2,7
print(x)
```

- a. 2
- b. 7
- c. (2,7)
- d. Error

What will be the output of the following code?

```
x=5.5
y=5.5
print(x+y)
```

- a. 11
- b. 11.0
- c. 10
- d. 0

What will be the output of the following code?

```
a=25.0
b=5.0
print(a%b)
```

- a. 5
- b. 5.0
- c. 0
- d. 0.0

What will be the output of the following code?

```
i=47
j=25
i=i%10
j=j%10
print(i*j)
```

- a. 8
- b. 63
- c. 35
- d. 72

Thu, 26 Mar | 88% | ↓ 0.01 MB/s
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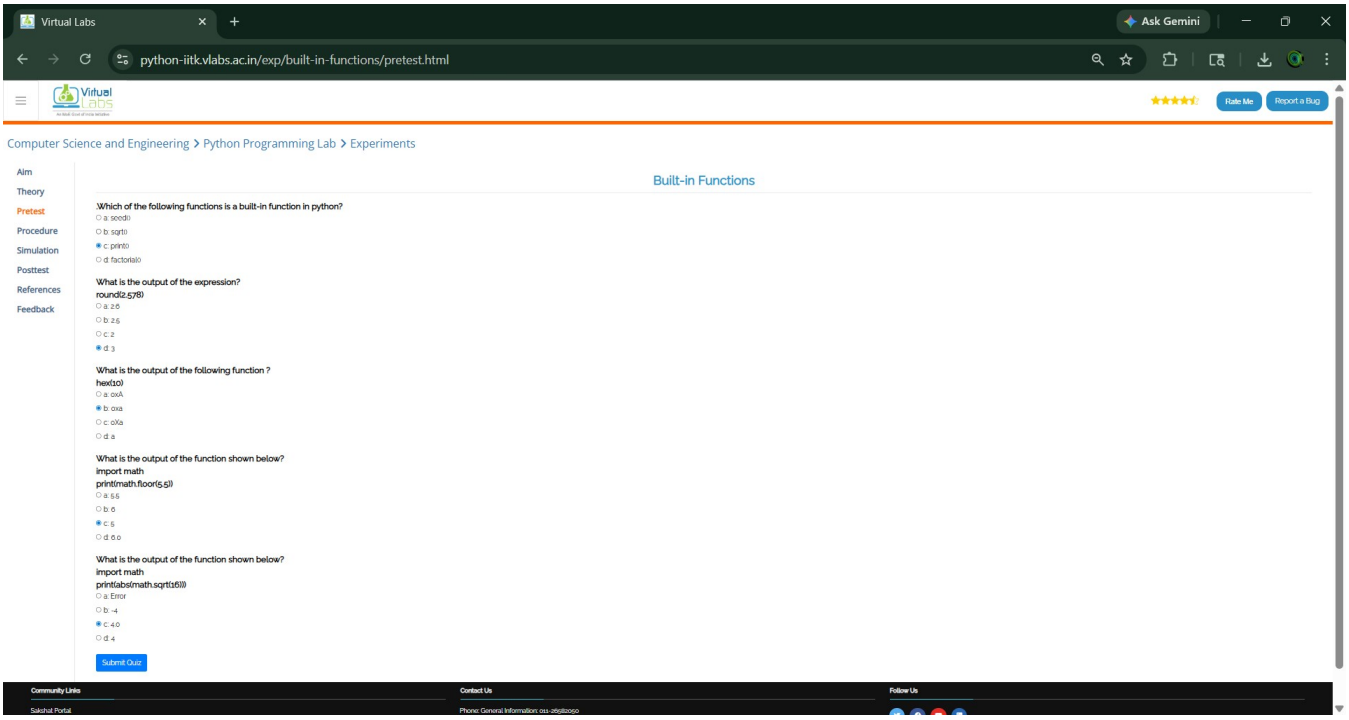
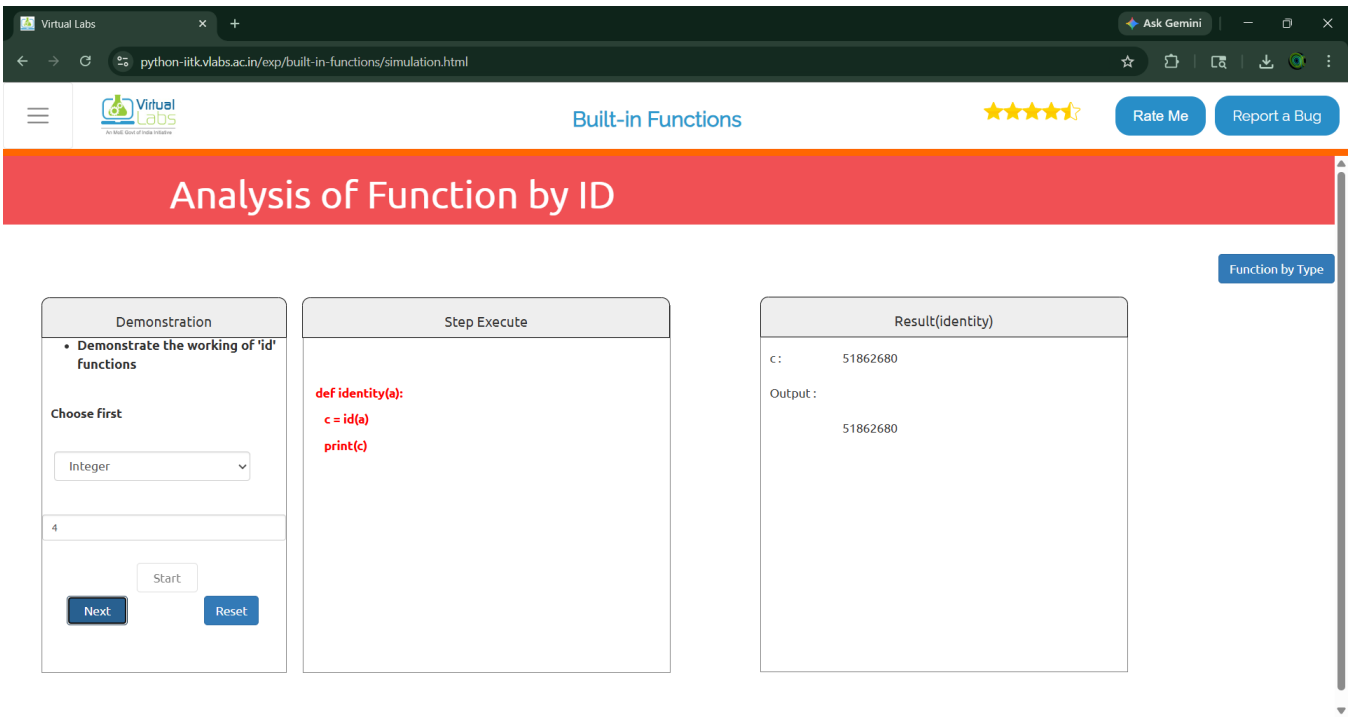
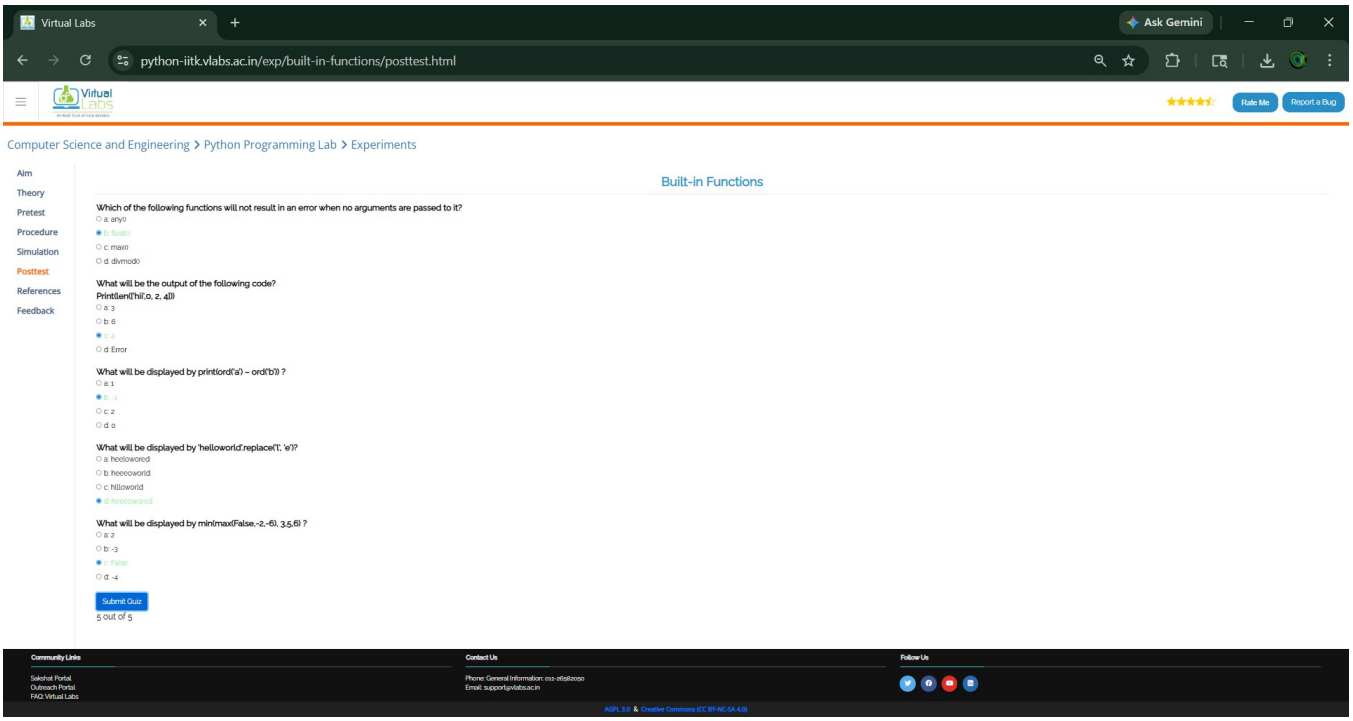
Built-in Functions



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Built-in Functions
Which of the following functions will not result in an error when no arguments are passed to it?
a: any()
b: float()
c: max()
d: divmod()
What will be the output of the following code?
Print(len('hi',0, 2, 4))
a: 3
b: 6
c: 4
d: Error
What will be displayed by print(ord('a') – ord('b')) ?
a: 1
b: -1
c: 2
d: 0
What will be displayed by 'helloworld'.replace('l', 'e')?
a: heelowored
b: heeeoworld
c: hlllloworld
d: heeeoworld
What will be displayed by min(max(False,-2,-6), 3,5,6) ?
a: 2
b: -3
c: False
d: -4
Submit Quiz
5 out of 5
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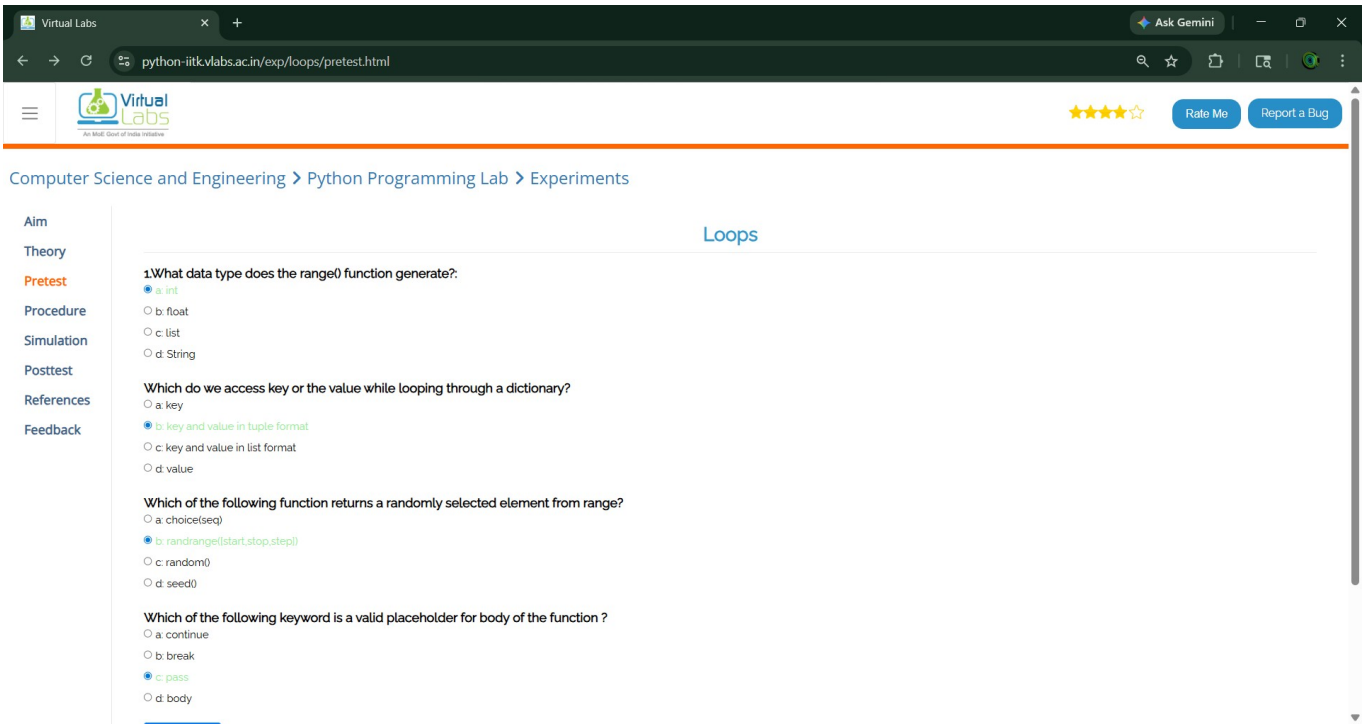
Loops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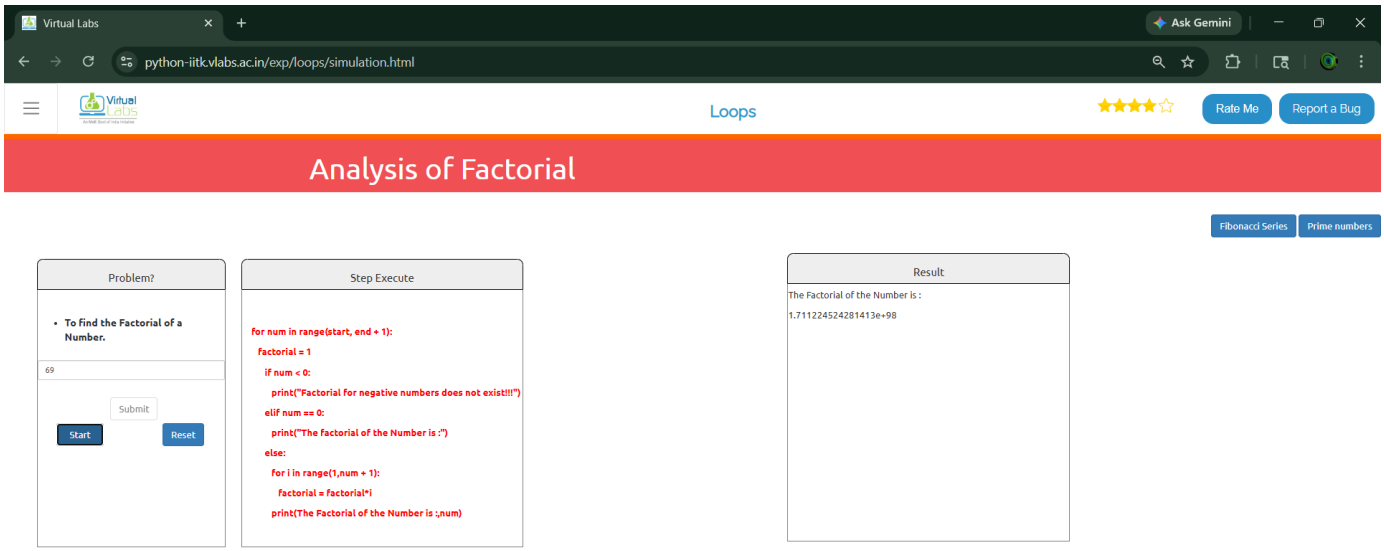
Loops
1.What data type does the range() function generate?:
a: int
b: float
c: list
d: String
Which do we access key or the value while looping through a dictionary?
a: key
b: key and value in tuple format
c: key and value in list format
d: value
Which of the following function returns a randomly selected element from range?
a: choice(seq)
b: randrange([start,stop,step])
c: random()
d: seed()
Which of the following keyword is a valid placeholder for body of the function ?
a: continue
b: break
c: pass
d: body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Loops
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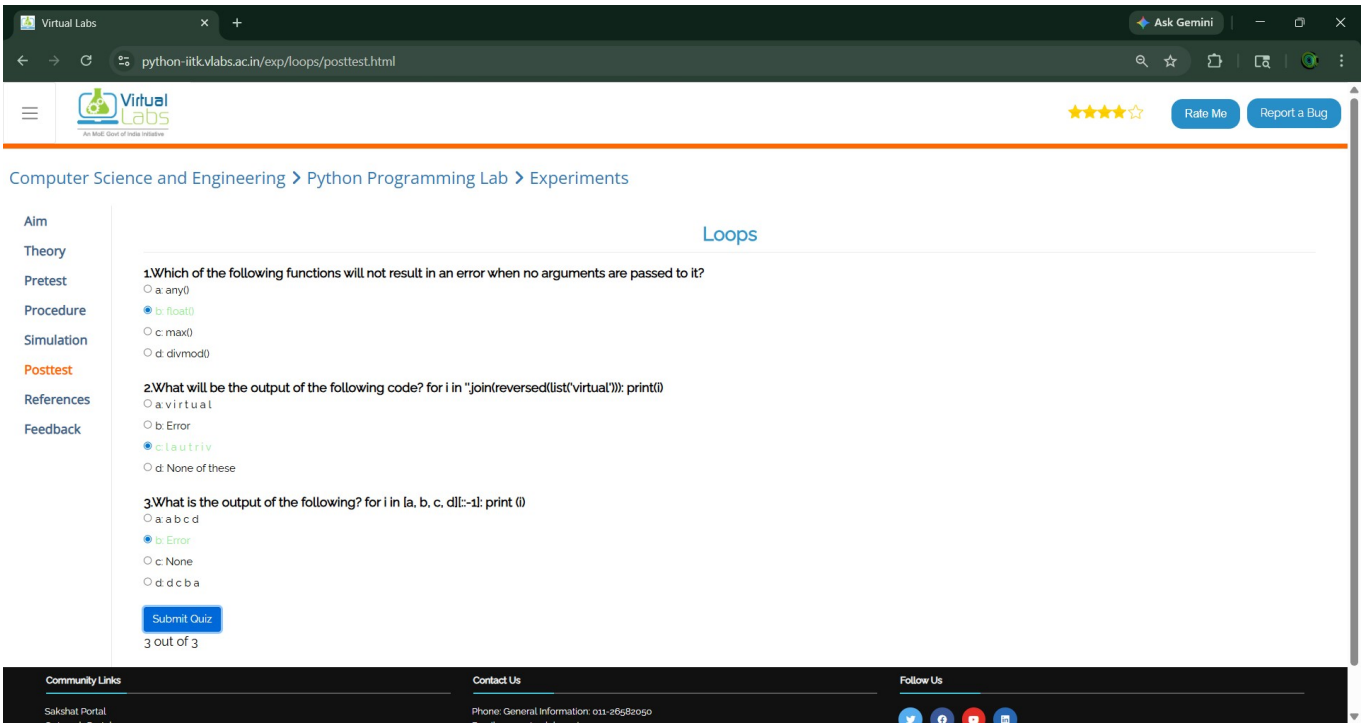
Analysis of Factorial
Fibonacci Series
Prime numbers
Problem?
To find the Factorial of a Number.
69
Submit
Start
Reset
Step Execute
for num in range(start, end + 1):
factorial = 1
if num < 0:
print("Factorial for negative numbers does not exist!!!")
elif num == 0:
print("The factorial of the Number is :")
else:
for i in range(1,num + 1):
factorial = factorial*i
print(The Factorial of the Number is :,num)
Result
The Factorial of the Number is :
1.711224524281413e+98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Loops
1.Which of the following functions will not result in an error when no arguments are passed to it?
a: any()
b: float()
c: max()
d: divmod()
2.What will be the output of the following code? for i in ''.join(reversed(list('virtual'))): print(i)
a: v i r t u a l
b: Error
c: l a u t r i v
d: None of these
3.What is the output of the following? for i in [a, b, c, d][::-1]: print (i)
a: a b c d
b: Error
c: None
d: d c b a
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3 out of 3
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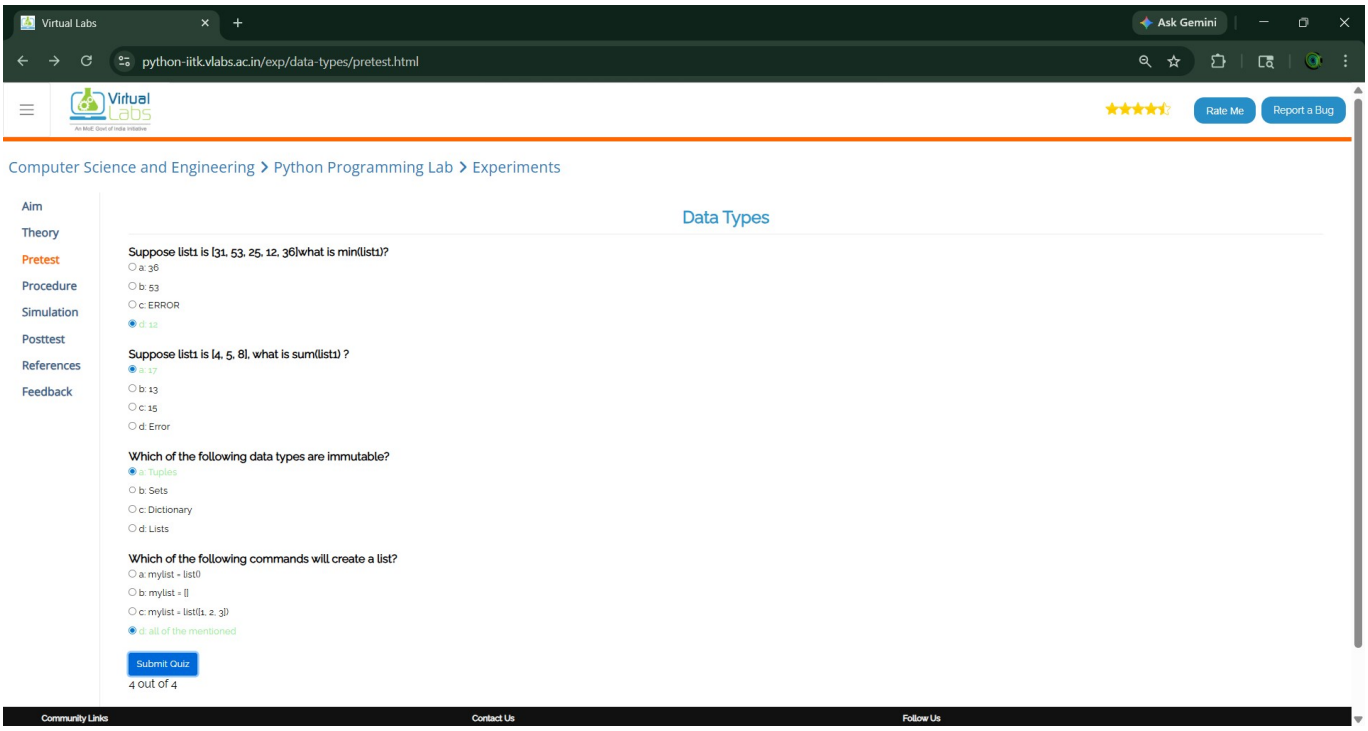
Data Types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Data Types
Suppose list1 is [31, 53, 25, 12, 36]what is min(list1)?
a: 36
b: 53
c: ERROR
d: 12
Suppose list1 is [4, 5, 8], what is sum(list1) ?
a: 17
b: 13
c: 15
d: Error
Which of the following data types are immutable?
a: Tuples
b: Sets
c: Dictionary
d: Lists
Which of the following commands will create a list?
a: mylist = list()
b: mylist = []
c: mylist = list([1, 2, 3])
d: all of the mentioned
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4 out of 4
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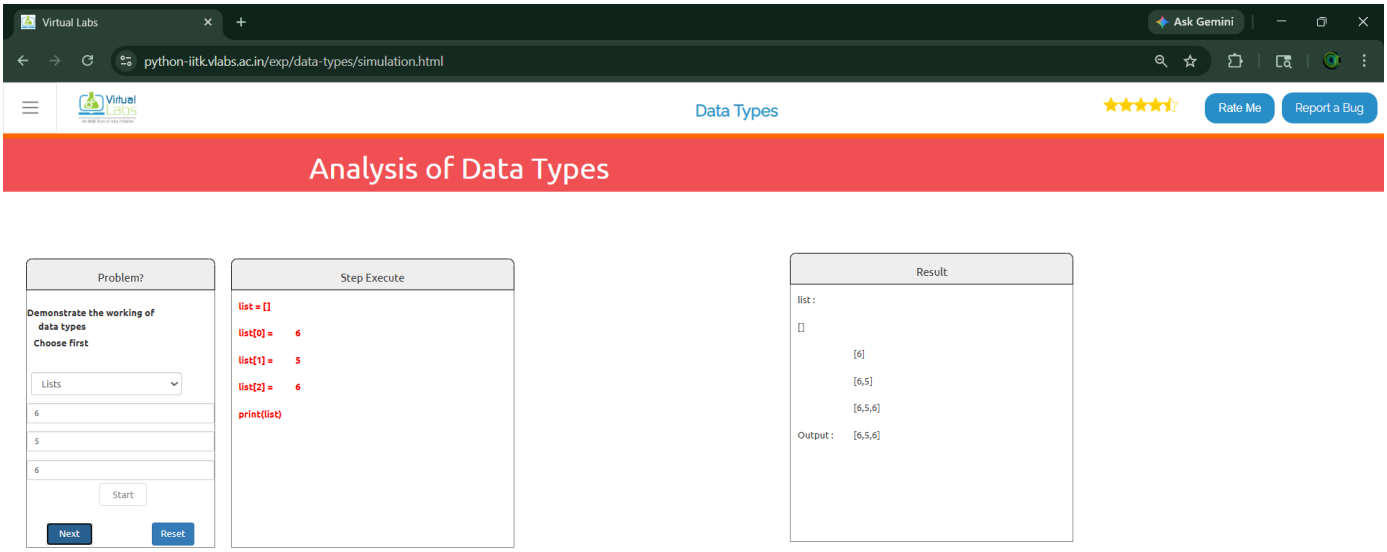
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Data Types
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Analysis of Data Types
Problem?
Demonstrate the working of data types
Choose first
Lists
6
5
6
Start
Next
Reset
Step Execute
list = []
list[0] = 6
list[1] = 5
list[2] = 6
print(list)
Result
list :
[]
[6]
[6,5]
[6,5,6]
Output : [6,5,6]

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Data Types

What will be the output of the following

```
fruits = ['apples', 'guava', 'bananas']
oranges = (4, 3)
fruits.append(oranges)
print(fruits)
```

- a: ['apples', 'guava', 'bananas', (4, 3)]
- b: Ledger can be viewed by anyone. ['apples', 'guava', 'bananas', 4, 3]
- c: ['apples', 'guava', 'bananas', (4, 3)
- d: ['apples', 'guava', 'bananas'], (4, 3)

If mylist is [0, 3, 2, 5] then what is mylist * 2?

- a: [0, 6, 4, 10]
- b: [0, 3, 2, 5, 0, 3]
- c: , [0, 3, 2, 5, 0, 3, 2, 5]
- d: [0, 3, 2, 5, 5, 3, 2, 1]

What will be the output of the following ?

list1 = [0, 2, 7, 4, 5]

print(list1[3: 0: -1])

- a: Syntax error
- b: [4, 7, 2]
- c: [4,7]
- d: [4, 7, 2,0]

What will be the output?

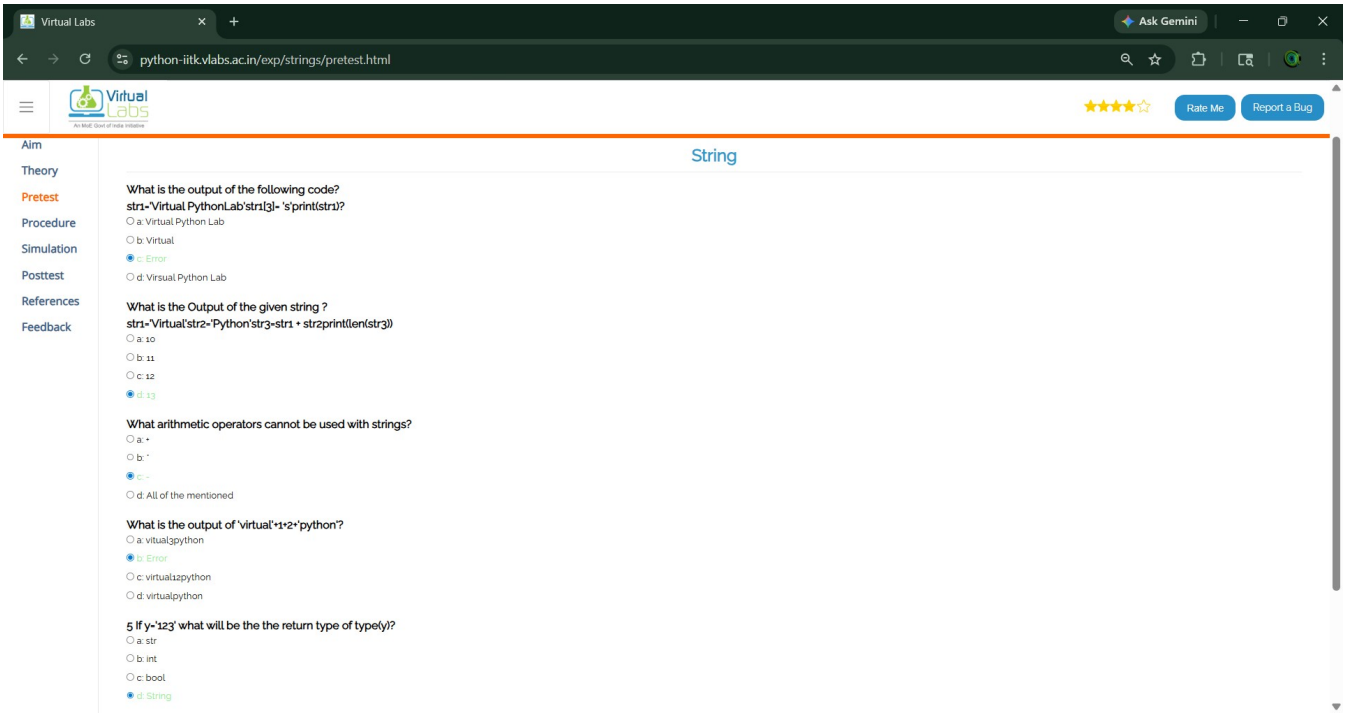
d1 =['james': 40, 'peter': 45]

d2 = { 'james': 466, 'peter': 45 }

d1 > d2

- a: True
- b: False
- c: Error
- d: None

String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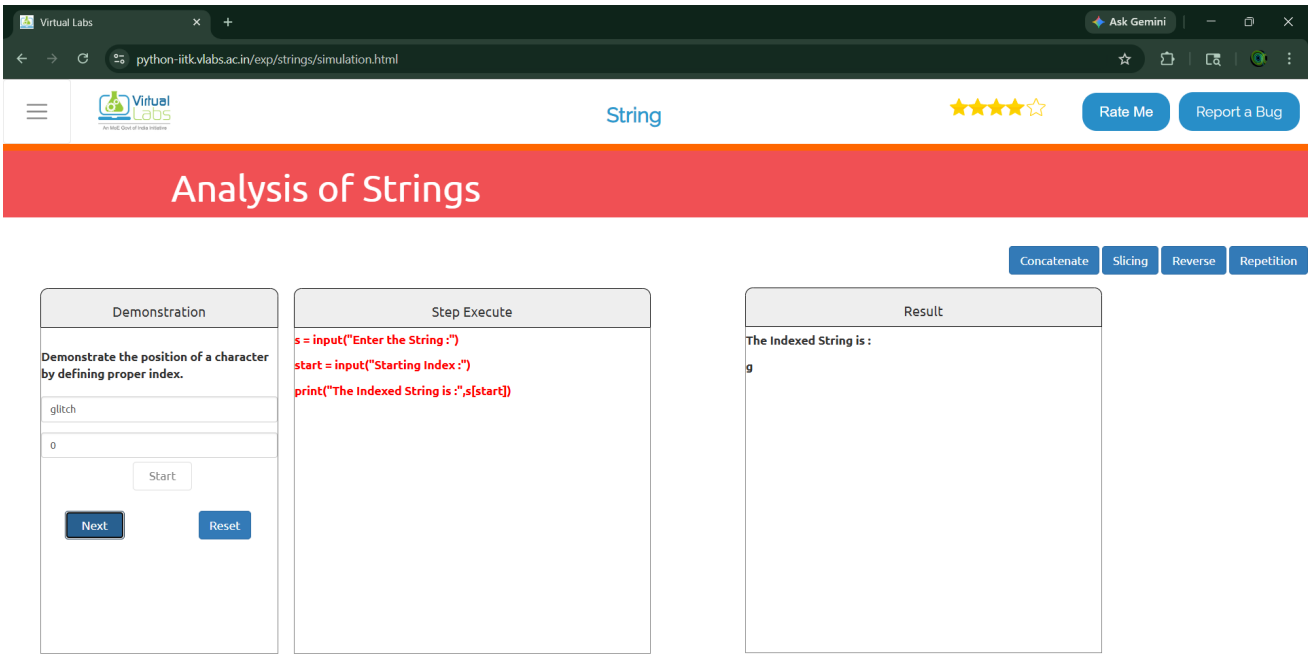
String
What is the output of the following code?
str1='Virtual PythonLab'str1[3]- 's'print(str1)?
a: Virtual Python Lab
b: Virtual
c: Error
d: Virsual Python Lab
What is the Output of the given string ?
str1='Virtual'str2='Python'str3=str1 + str2print(len(str3))
a: 10
b: 11
c: 12
d: 13
What arithmetic operators cannot be used with strings?
a: +
b: *
c: -
d: All of the mentioned
What is the output of 'virtual'+1+2+'python'?
a: vitual3python
b: Error
c: virtual12python
d: virtualpython
5 If y='123' what will be the the return type of type(y)?
a: str
b: int
c: bool
d: String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String
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Analysis of Strings
Concatenate
Slicing
Reverse
Repetition
Demonstration
Demonstrate the position of a character by defining proper index.
glitch
0
Start
Next
Reset
Step Execute
s = input("Enter the String :")
start = input("Starting Index :")
print("The Indexed String is :",s[start])
Result
The Indexed String is :
g

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String

What is the output of the following code

```
class customer:
    def __init__(self, id):
        self.id = str(id)

id = '2010'
obj = customer(2567)
print(obj.id)
```

- a: 2567
- b: Error
- c: 2010
- d: None

What is the output when the following code is executed?
str1='virtualpythonla'

- a: Simsudin
- b: hello
- c: world
- d: helloworld

What is the output of the following Python program

```
text = 'Python is high-level programming language'
words = text.split()
l = len(words)
print('Number of words of the text: %d' % l)
```

- a: Error
- b: 5
- c: 6
- d: None of these

What is the output of the following code

```
def changement(str1):
    char = str1[0]
    str1 = str1[0] + str1[1:].replace(char, '&')
    return str1

print(changement('this is the string'))
```

- a: Syntax Error
- b: this is the string
- c: &his is &he s&tring
- d: this is &he s&tring

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STUDENT DECLARATION

I hereby declare that the Virtual Lab experiments included in this document have been performed by me as part of my academic activity. The screenshots submitted are genuine and taken during the actual execution of the experiments.

Student Signature:

Date: 26 March 2026